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United States Patent	12409087
Kind Code	B2
Date of Patent	September 09, 2025
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User controls for patient support apparatus having low height

Abstract

A patient support apparatus comprises a support structure. The support structure comprises a base and a patient support deck coupled to the base. The patient support deck comprises a patient support surface capable of articulating to adjust positioning of a patient supported thereon, a lift system configured to adjust a height of the patient support deck relative to the base, and a user interface system coupled to the support structure and comprising an elongated member and a user interface coupled to the elongated member. The user interface is configured to receive inputs from a user to control a function of the patient support apparatus.

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Appl. No.: 17/125058

Filed: December 17, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210100705 A1	Apr. 08, 2021

Related U.S. Application Data

continuation parent-doc US 16210872 20181205 US 10898399 child-doc US 17125058
us-provisional-application US 62607956 20171220

Publication Classification

Int. Cl.: **A61G7/018** (20060101); **A61B5/00** (20060101); **A61G7/012** (20060101); **A61G7/05** (20060101); **A61G7/08** (20060101); **A61G7/10** (20060101)

U.S. Cl.:

CPC **A61G7/018** (20130101); **A61G7/012** (20130101); **A61G7/0528** (20161101); **A61G7/08** (20130101); **A61G7/1013** (20130101); A61B5/6892 (20130101); A61B2562/0252 (20130101); A61G7/0506 (20130101); A61G7/051 (20161101); A61G7/0514 (20161101); A61G7/0524 (20161101); A61G7/1046 (20130101); A61G7/1067 (20130101); A61G2203/32 (20130101); A61G2203/36 (20130101); A61G2203/40 (20130101); A61G2203/42 (20130101)

Field of Classification Search

CPC: A61G (7/012); A61G (7/018); A61G (7/0506); A61G (7/051); A61G (7/054); A61G (7/0514); A61G (7/0524); A61G (7/0528); A61G (7/08); A61G (7/1013); A61G (7/1046); A61G (7/1067); A61G (2203/32); A61G (2203/36); A61G (2203/40); A61G (2203/42); A61B (5/6892); A61B (2562/0252); A61B (2017/00207); G16H (40/60)

USPC: 5/613

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The subject patent application is a Continuation of U.S. patent application Ser. No. 16/210,872, filed on Dec. 5, 2018 which claims priority to and all the benefits of U.S. Provisional Patent Application No. 62/607,956 filed on Dec. 20, 2017, the disclosures of each of which are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Patient transport systems facilitate care of patients in a health care setting. Patient transport systems comprise patient support apparatuses (e.g., beds, stretchers, cots, tables, wheelchairs, chairs, recliners, etc.). Patient support apparatuses comprise a base, a support frame upon which the patient is supported, a lift system for lifting and lowering the support frame relative to the base, and a control interface for controlling the lift system. Because the lift system lifts and lowers the support frame, the control interface of the lift system must be accessible to a caregiver at different heights. However, the control interface may be difficult to access when it is located on the support frame itself, due to the varying heights of the support frame. For example, when the support frame is at a low height, for purposes such as fall prevention, the control interface may be too low for the caregiver to reach easily or ergonomically.

(2) A patient support apparatus having user controls designed to overcome at least one or more of the aforementioned disadvantages is desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is perspective view of a patient support apparatus according to one example.

(2) FIG. 2 is a perspective view of a lift system of the patient support apparatus at a maximum height.

(3) FIG. 3 is a perspective view of the lift system of the patient support apparatus at a minimum height.

(4) FIG. 4 is a block diagram of one example of a control system for the patient support apparatus.

(5) FIG. 5 is a front view of a user interface system for the patient support apparatus.

(6) FIG. 6 is a flowchart of a method of operating the control system according to one example.

DETAILED DESCRIPTION

(7) I. Patient Support Apparatus Overview

(8) Referring to FIG. 1, a patient support apparatus **30** is shown for supporting a patient in a health care setting. The patient support apparatus **30** illustrated in FIG. 1 comprises a hospital bed. In other embodiments, however, the patient support apparatus **30** may comprise a stretcher, cot, table, wheelchair, chair, or similar apparatus utilized in the care of a patient.

(9) A support structure **32** defines various components of the chassis and/or body of the patient support apparatus **30**. Often, the support structure **32** provides support for the patient. The support structure **32** illustrated in FIG. 1 comprises a base **34** and a support frame **36**. The base **34** comprises a base frame **35**. The support frame **36** is spaced above the base frame **35** in FIG. 1. The

support structure **32** also comprises a patient support deck **38** disposed on the support frame **36**. The patient support deck **38** comprises several sections, some of which are pivotable relative to the support frame **36**, such as a fowler section, a seat section, a thigh section, and a foot section. The patient support deck **38** provides a patient support surface **42** upon which the patient is supported.

(10) A mattress (not shown) is disposed on the patient support deck **38** during use. The mattress comprises a secondary patient support surface upon which the patient is supported. The base **34**, support frame **36**, patient support deck **38**, and patient support surfaces **42** each have a head end and a foot end corresponding to designated placement of the patient's head and feet on the patient support apparatus **30**. The base **34** comprises a longitudinal axis L along its length from the head end to the foot end. The base **34** also comprises a vertical axis V arranged crosswise (e.g., perpendicularly) to the longitudinal axis L along which the support frame **36** is lifted and lowered relative to the base **34**. The construction of the support structure **32** may take on any design not limited to that specifically set forth above. In addition, the mattress may be omitted in certain embodiments, such that the patient rests directly on the patient support surface **42**.

(11) The support structure **32** comprises side rails **44, 46, 48, 50** coupled to the support frame **36** and thereby supported by the base **34**. A first side rail **44** is positioned at a right head end of the support frame **36**. A second side rail **46** is positioned at a right foot end of the support frame **36**. A third side rail **48** is positioned at a left head end of the support frame **36**. A fourth side rail **50** is positioned at a left foot end of the support frame **36**. If the patient support apparatus **30** is a stretcher or a cot, there may be fewer side rails. The side rails **44, 46, 48, 50** are movable between a raised position in which they block ingress and egress into and out of the patient support apparatus **30**, one or more intermediate positions, and a lowered position in which they are not an obstacle to such ingress and egress. In still other configurations, the patient support apparatus **30** may not include any side rails.

(12) The support structure **32** comprises a headboard **52** and a footboard **54**, which are coupled to the support frame **36**. In other embodiments, when the headboard **52** and footboard **54** are included, the headboard **52** and footboard **54** may be coupled to other locations on the patient support apparatus **30**, such as the base **34**. In still other embodiments, the patient support apparatus **30** does not include the headboard **52** and/or the footboard **54**.

(13) Caregiver interfaces **56**, such as handles, are shown integrated into the footboard **54** and side rails **44, 46, 48, 50** to facilitate movement of the patient support apparatus **30** over floor surfaces. Additional caregiver interfaces **56** may be integrated into the headboard **52** and/or other components of the patient support apparatus **30**. The caregiver interfaces **56** are graspable by the caregiver to manipulate the patient support apparatus **30** for movement.

(14) Other forms of the caregiver interface **56** are also contemplated. The caregiver interface may comprise one or more handles coupled to the support frame **36**. The caregiver interface may simply be a surface on the patient support apparatus **30** upon which the caregiver logically applies force to cause movement of the patient support apparatus **30** in one or more directions, also referred to as a push location. This may comprise one or more surfaces on the support frame **36** or base **34**. This could also comprise one or more surfaces on or adjacent to the headboard **52**, footboard **54**, and/or side rails **44, 46, 48, 50**. In other embodiments, the caregiver interface may comprise separate handles for each hand of the caregiver. For example, the caregiver interface may comprise two handles.

(15) A drive system **90** influences motion of the patient support apparatus **30** during transportation over a floor surface. The drive system **90** may have one or more motors (not shown), or may be non-powered. The drive system **90** is integrated into the patient support apparatus **30**. The drive system **90** may comprise wheels **58**, which are coupled to the base **34** to facilitate transport over the floor surfaces. The wheels **58** are arranged in each of four quadrants of the base **34** adjacent to corners of the base **34**. In the embodiment shown, the wheels **58** are caster wheels able to rotate and swivel relative to the support structure **32** during transport. Each of the wheels **58** forms part of a

caster assembly **60**. Each caster assembly **60** is mounted to the base **34**. It should be understood that various configurations of the caster assemblies **60** are contemplated. In addition, in some embodiments, the wheels **58** are not caster wheels and may be non-steerable, steerable, non-powered, powered, or combinations thereof. Additional wheels are also contemplated. For example, the patient support apparatus **30** may comprise four non-powered, non-steerable wheels, along with one or more powered wheels. In some cases, the patient support apparatus **30** may not include any wheels.

(16) In other embodiments, one or more auxiliary wheels (powered or non-powered), which are movable between stowed positions and deployed positions, may be coupled to the support structure **32**. In some cases, when these auxiliary wheels are located between caster assemblies **60** and contact the floor surface in the deployed position, they cause two of the caster assemblies **60** to be lifted off the floor surface thereby shortening a wheel base of the patient support apparatus **30**. A fifth wheel may also be arranged substantially in a center of the base **34**.

(17) Further details of a powered drive system that may be used with the patient support apparatus **30** disclosed herein are disclosed in U.S. Pat. No. 6,772,850, issued to Waters et al. and entitled POWER ASSISTED WHEELED CARRIAGES, as well as U.S. Pat. No. 9,259,369, issued to Derenne et al. and entitled POWERED PATIENT SUPPORT APPARATUS, the complete disclosures of which are both hereby incorporated herein by reference.

(18) II. User Control Techniques for Low Height

(19) As shown in FIG. 1, a user interface system **62** is coupled to the support structure **32**. The user interface system **62** may comprise an elongated member **64**. The elongated member **64** may comprise a vertically extending member, i.e., extending upwards along axis V. The elongated member **64**, in one embodiment, is a pole, and more specifically, may be an IV pole.

(20) The user interface system **62** further comprises a user interface **66** coupled to the elongated member **64**. The user interface **66** may be configured to receive inputs from a user to control one or more functions of the patient support apparatus **30**.

(21) The elongated member **64** as shown in FIG. 2 has a cylindrical cross-section. However, the elongated member may have any other cross-section, such as rectangular, square, etc. The elongated member **64** has a height defined between a proximal end, which is coupled to the support structure **32** and a distal end, which is freely positioned in space. As will be understood from the Figures and the examples described herein, the height of the elongated member **64** is specifically defined to enable the user interface **66** to remain at an ergonomically suitable position whether the support frame **36** of the patient support apparatus **30** is lifted to a maximum height or lowered to a minimum height.

(22) As shown in FIG. 2, the patient support apparatus **30** may further comprise a second user interface **68** coupled to the support structure **32**. In various embodiments, the second user interface **68** may be physically coupled to, for example, the footboard **52**, the side rails **44**, **46**, **48**, **50**, or any other suitable placement on the support structure **32**. The second user interface **68** may be configured to receive inputs from a user to control one or more functions of the patient support apparatus **30**.

(23) Referring to FIGS. 2 and 3, the patient support apparatus **30** comprises a lift system **70** that operates to lift and lower the support frame **36** relative to the base **34**. The lift system **70** is configured to move the support frame **36** from the minimum height (shown in FIG. 3) to the maximum height (shown in FIG. 2), or to any desired position in between.

(24) In one example, as shown, the lift system **70** comprises head end and foot end lift members **72**, **74**. First and second actuators **73**, **75** move the lift members **72**, **74** to lift and lower the support frame **36** relative to the base **34**. The first actuator **73** is coupled to the head end lift member **72**. The second actuator **75** is coupled to the foot end lift member **74**. The actuators **73**, **75** operate to pivot their respective lift member **72**, **74** about fixed upper pivot axes P2 to lift and lower the support frame **36** relative to the base **34**, as described further below.

(25) The actuators **73, 75** comprise linear actuators, rotary actuators, or other types of actuators. The actuators **73, 75** may be electrically operated and/or may be hydraulic. In the embodiment shown, the actuators **73, 75** are electro-hydraulic, linear actuators, such as compact electro-hydraulic actuators. In other embodiments, the actuators **73, 75** can be electric, linear actuators. It is contemplated that, in some embodiments, only one lift member and one associated actuator may be employed, e.g., to raise only one end of the support frame **36**.

(26) The lift members **72, 74** comprise a pair of head end lift legs **76** and a pair of foot end lift legs **78** pivoted by the actuators **73, 75** about the fixed upper pivot axes **P2**. In other embodiments, each of the lift members **72, 74** may comprise a single lift leg. In still other embodiments, other types of lifting members capable of lifting and lowering the support frame **36** may be employed. The lift members **72, 74** may be identical in form or may have different forms. For instance, one of the lift members **72, 74** may be a single lift leg, while the other of the lift members **72, 74** may comprise part of a scissor-type mechanism. It should be appreciated that each of the lift members **72, 74** may be formed in a unitary construction or may be separate pieces fastened together.

(27) The lift members **72, 74** comprise first end sections **80, 82** movably coupled to the base **34**. In particular, the first end sections **80, 82** are connected to guided bodies **108** that slide in head end and foot end guides **100, 102** relative to the base **34** during the lifting and lowering of the support frame **36**, i.e., when the actuators **73, 75** pivot the lift members **72, 74** about the fixed upper pivot axes **P2**. In the embodiment shown, the first end sections **80, 82** comprise first ends of the lift legs **76, 78** and a support member **83, 85** interconnecting each pair of the lift legs **76, 78**, respectively, at their first ends. In the embodiment shown, the support members **83, 85** are rigidly fixed to the lift legs **76, 78** to move with the lift legs **76, 78**. The support members **83, 85** define a moving lower pivot axis **P1** about which the support members **83, 85** pivot as the first end sections **80, 82** slide relative to the base **34**. In other embodiments, the lift legs **76, 78** may pivot relative to the support members **83, 85**.

(28) The lift members **72, 74** extend from the first end sections **80, 82** to second end sections **84, 86**. The second end sections **84, 86** are pivotally connected to the support frame **36** at the fixed upper pivot axes **P2** for pivoting relative to the support frame **36**. In the embodiment shown, the second end sections **84, 86** comprise second ends of the lift legs **76, 78**. The fixed upper pivot axes **P2** lie in a common plane perpendicular to the vertical direction when the support frame **36** is at the minimum height or the maximum height.

(29) The guides **100, 102** are arranged to guide the movement of the first end sections **80, 82** when the actuators **73, 75** pivot the lift members **72, 74** about the fixed upper pivot axes **P2** to lift and lower the support frame **36** relative to the base **34**. The head end guides **100** guide movement of the head end lift member **72**. The foot end guides **102** guide movement of the foot end lift member **74**. In the embodiment shown, four guides **100, 102** are provided. The four guides **100, 102** comprise a pair of head end guide tracks **104** and a pair of foot end guide tracks **106**. The guide tracks **104, 106** are fixed to the base **34** and have a hollow, elongated shape. In particular, the guide tracks **104, 106** are shown being formed of rectangular tubing. In other embodiments, the guides **100, 102** may assume other forms or shapes capable of guiding movement of the first end sections **80, 82** of the lift members **72, 74**.

(30) While the lift system **70**, as shown in FIGS. 2 and 3, has been described in detail, those skilled in the art appreciate that the lift system **70** may be implemented according to various other different configurations beside that shown in FIGS. 2 and 3.

(31) A height sensor **92** may be coupled to the support frame **36** (as shown in FIG. 2), the patient support deck **38**, or any other suitable location. The height sensor **92** may detect a height of the patient support deck **38** relative to the base **34**. The height sensor **92** may comprise a 3-D accelerometer, a 3-D gyroscope, an inclinometer, or other suitable type of sensor configured to directly or indirectly measure height changes.

(32) Referring now to FIG. 4, a block diagram of a control system **94** usable with the patient

support apparatus **30** is shown, according to one embodiment. As described, when the patient support deck **38** is at a low height, it is desirable for the user controls to be located at a higher level that is more easily accessible to the user. Therefore, when the lift system **70** lowers the patient support deck **38** relative to the base **34**, the height sensor **92** may detect that the height of the patient support deck **38** is below a predefined threshold or at the minimum height. In other examples, the height sensor **92** may identify that the support deck **38** is in the process of being lowered.

(33) The height sensor **92** may indicate such readings to a controller **96** of the patient support apparatus **30**. The controller **96** may be in communication with and may control any suitable components of the patient support apparatus **30**, such as the electrical or electromechanical components described herein. The controller **96** may comprise any suitable signal processing means, computer executable instructions or software modules stored in a non-transitory memory wherein the executable instructions or modules may be executed by a processor, or the like. Additionally, or alternatively, the controller **96** may comprise a microcontroller, a processor, one or more integrated circuits, logic parts, and the like for enabling the same. The controller **96** may have any suitable configuration for enabling performance of various tasks related to operation of the patient support apparatus **30**, such as those described below. The controller **96** may be located at any suitable location of the patient support apparatus **30**.

(34) In other examples, the controller **96** may determine that the support deck **38** is at a low/minimum height based on input from sensors besides the height sensor **92**. For instance, the controller **96** may receive readings from the any of the moveable components of the lift system **70**, as described above. Additionally, the controller **96** may infer the height of the support deck **38** after executing height adjustment commands triggered from any of the user interfaces **66**, **68**.

(35) The user interface **66** coupled to the elongated member **64** may be activated by the controller **96** upon the controller **96** determining that the support deck **38** is at a low/minimum height. Therefore, even when the patient support deck **38** is at a low height, controls may be ergonomically accessible to the user.

(36) When the user interface **66** is activated, the user interface **66** is configured to receive input from the user. The user interface **66** may generate an input signal based on the received input and send the input signal to the controller **96**. The user interface **66** may receive inputs from the user to control operation of the lift system **70** or to control operation of one or more other powered devices on the patient support apparatus **30** to one or more of: (1) adjust the height of the patient support deck **38** relative to the base **34**; adjust a tilt of the patient support deck **38** relative to the base **34**; (3) control articulation of the patient support surface **38**; (4) control movement of the patient support apparatus **30** over a floor surface; (5) control operation of a patient scale; and (6) control electronic brakes or other mobility system. The user interface **66** may be used to control operation of any powered device associated with the patient support apparatus **30**.

(37) In one embodiment, in addition to or instead of receiving inputs from the user to control operation of the lift system **70**, the user interface **66** may receive inputs from the user to activate a bed exit detection system to generate an alarm in certain situations. For example, the alarm may indicate that a patient has exited the patient support apparatus **30**.

(38) When the lift system **70** raises the patient support deck **38**, it may be desirable for the user controls to be located at a lower level that is more easily accessible to the user. Therefore, when the lift system **70** raises the patient support deck **38** relative to the base **34**, the height sensor **92** may detect that the height of the patient support deck **38** is above the predefined threshold or at the maximum height. The height sensor **92** may indicate to the controller **96** that the patient support deck **38** is above the predefined threshold or at the maximum height. In other examples, the height sensor **92** may identify that the support deck **38** is in the process of being raised. In other examples, the controller **96** may determine that the support deck **38** is at a high/maximum height based on input from sensors besides the height sensor **92**. For instance, the controller **96** may receive

readings from the any of the moveable components of the lift system **70**, as described above.

Additionally, the controller **96** may infer the height of the support deck **38** after executing height adjustment commands triggered from any of the user interfaces **66**, **68**.

(39) In such instances, the second user interface **68**, which is disposed at a lower height relative to the user interface **66** coupled to the elongated member **64**, may be activated by the controller **96** upon the controller **96** determining that the support deck **38** is at a high/maximum height. Therefore, even when the patient support deck **38** is at an elevated height, controls may be ergonomically accessible to the user.

(40) When the second user interface **68** is activated, the second user interface **68** is configured to receive input from the user. The second user interface **68** may generate an input signal based on the received input and send the input signal to the controller **96**. The second user interface **68** may receive inputs from the user to control operation of any of the aforementioned capabilities described with respect to the user interface **66**.

(41) In one embodiment, both the user interface **66** and the second user interface **68** are simultaneously activated, such that the user may choose to use the interface most conveniently accessible to the user based on the user's position and the position of the patient support deck **38**. In an alternate embodiment, only one of the user interface **66** and the second user interface **68** is activated at a time, based on the position of the patient support deck **38**. In yet another embodiment, the second user interface **68** is always activated, and the user interface **66** is activated only when the height of the patient support deck **38** is below the predefined threshold.

(42) Referring now to FIG. 5, a front view of one example of the user interface system **62** is shown. The user interface **66** may comprise a movable interface configured to be translated along, tilted relative to, and/or rotated about the elongated member **64**. For instance, in the illustrated embodiment, the user interface **66** may be a slidable input device configured to be translated along the elongated member **64** by the user. The arrows indicate upward and downward movement of the user interface **66** along the elongated member **64**. In one version a spring may couple the user interface **66** to the elongated member **64** so that movement of the user interface **66** in either direction is returned to a neutral, home position. In this case, movement of the user interface **66** may engage one or more switches or other sensors when moved a predetermined distance in either direction.

(43) In embodiments where the user interface system **62** comprises more than one movable interface **66**, each movable interface **66** may be separately movable relative to the elongated member **64**. In an alternate embodiment, the user interface **66** may be tiltable relative to a vertical axis of the elongated member **64**. In another alternate embodiment, the user interface **66** may be a rotatable input device configured to be rotated about the elongated member **64** by the user. Combinations of various types of user interfaces **66** are also contemplated.

(44) In one embodiment, the user interface **66** comprises a capacitive sensor. For example, capacitive sensors can be integrated into the elongated member **64** so that they are substantially flush with an outer surface of the elongated member to provide an essentially smooth outer surface. Such configuration provides a sanitary and easily cleanable surface suitable for hospital environments whereby the flush surface is free of any grooves thereby preventing accumulation of biomass or bacteria.

(45) Capacitive sensors may also be raised slightly to provide some tactile identification to the user. Separate capacitive sensors can be employed to sense swiping with a user's finger or other body part to move parts of the patient support apparatus **30** in each direction—for example one capacitive sensor senses the user's finger to lift the support frame **36** and one capacitive sensor senses the user's finger to lower the support frame **36**. In other embodiments, the user interface **66** comprises piezoelectric sensors coupled to the elongated member **64** to be actuated by the user, e.g., by contact with a user's finger. In this case, one piezoelectric sensor may sense the user's finger to lift the support frame **36** and one piezoelectric sensor may sense the user's finger to lower

the support frame **36**.

(46) Referring again to FIG. **4**, the user interface system **62** may be coupled to the controller **96**. The controller **96** may determine a control attribute associated with the user interface **66**. The control attribute may be one or more of: (1) force applied to the user interface **66** (e.g., sensed by spring tension or compression or sensed by pressure on the piezoelectric sensor or other load cell or pressure sensor), (2) displacement of the user interface **66** (e.g., sliding, tilting, or rotating as can be measured by a potentiometer), (3) velocity of the user interface **66** (e.g., of a slidable, tiltable, or rotatable interface, or velocity of a finger sliding on a capacitive sensor), and/or (4) acceleration of the user interface **66** (e.g., of a slidable, tiltable, or rotatable interface as measured by an accelerometer, or acceleration of a finger sliding on a capacitive sensor).

(47) The controller **96** may operate the lift system **70**, drive system **90**, or other powered component of the patient support apparatus **30** based on the control attribute determined the controller **96** (by virtue of being coupled to one or more of the sensors, etc. described above). The operation of the lift system **70** may include one or more of: (1) displacement of the patient support deck **38** relative to the base **34**, (2) velocity in moving the patient support deck **38**, and (3) acceleration in moving the patient support deck **38**. In other words, the operation of the lift system **70** may be proportional to or have some other predetermined relationship to the control attribute (e.g., force, displacement, velocity, acceleration). For example, the power applied to the actuators **73**, **75** may be increased with an increase in the applied force.

(48) The controller **96** receives input signals from the one or more user interfaces **66**, **68** and can control operation of the actuator(s) **73**, **75** so that the user is given the impression of moving a weightless part of the patient support apparatus **30**. For example, the power applied to the actuators **73**, **75** by the controller **96** can be tuned so that the amount of upward sliding (or velocity or acceleration) of the user interface **66** associated with lift/lower is proportional to the amount of lifting (or lowering in the other direction).

(49) In one embodiment, as shown in FIGS. **4** and **5**, the user interface **66** may further comprise one or more indicators **98** for indicating to the user that the user interface **66** is activated to perform the function of the patient support apparatus **30**. The indicator **98** may be one or more of a visual indicator, a tactile indicator, and an audible indicator.

(50) In one embodiment, the indicator **98** is a light ring surrounding the user interface **66**, operated by an LED embedded within the user interface **66** (see FIG. **5**). When a user interface **66** is activated by the controller **96** based on the detected height of the patient support deck **38**, as described above, the LED turns the indicator **98** on, showing the activated state of the user interface **66** and indicating that the user may operate the user interface **66** to perform a function. When a user interface **66** is inactive, the LED turns the indicator **98** off, showing the inactive state of the user interface **66** and indicating that the user interface **66** is currently inoperable by the user due to the detected height of the patient support deck **38**.

(51) In other examples, the indicator **98** may indicate to the user information regarding an operation being executed from the user interface **66**. For example, the indicator **98** may illuminate an upward arrow to indicate that a component is being raised using the user interface **66** and illuminate a downward arrow to indicate that a component is being lowered using the user interface **66**.

(52) The user interface **66** provides a quick and intuitive means of operating the lift system **70**. As previously mentioned, placement of the user interface **66** at a higher level (e.g., on an IV pole mounted to the support frame **36**), allows the user to easily access the controls. Moreover, use of a slidable or rotatable user interface **66** may provide coarse or fine motion control, depending on the desired level of control. The simple operation of the user interface **66** may be preferable over a complex user interface, for example, a menu-based interface with multiple screens or an interface with multiple buttons/options, which a user must navigate to find and indicate the desired function and which may be difficult to operate at low heights.

(53) Various forms of the user interfaces **66**, **68** are contemplated for being actuated by the user,

such as the caregiver or the patient. Each user interface **66, 68** may be configured to be actuated in a variety of different ways, including but not limited to, mechanical actuation (hand, foot, finger, etc.), hands-free actuation (voice, foot, etc.), and the like. Each user interface **66, 68** may comprise a button, a gesture sensing device for monitoring motion of hands, feet, or other body parts of the caregiver (such as through a camera), a microphone for receiving voice activation commands, a foot pedal, and a sensor (e.g., infrared sensor such as a light bar or light beam to sense a user's body part, ultrasonic sensor, etc.). Additionally, the buttons/pedals can be physical buttons/pedals or virtually implemented buttons/pedals such as through optical projection (heads-up display) or on a touchscreen. It should be appreciated that any combination of user interfaces may also be utilized. The user interfaces **66, 68** may also be located on a portable electronic device (e.g., iWatch®, iPhone®, iPad®, or similar electronic devices or any other remote device/station in addition to a portable electronic device).

(54) Referring to FIG. **6**, a method of controlling movement of the patient support deck **38** relative to the base **34** is shown. In step **200**, the method starts by first detecting a height of the patient support deck **38**, i.e., by using the controller **96**.

(55) In a step **202**, the user interface **66** is activated by the controller **96** to generate an input signal upon receiving input from a user to control an operation of the patient support apparatus **30** when a detected height of the patient support deck **38** is below a predefined threshold or at the minimum height.

(56) In a step **204**, the user interface **66** receives inputs from the user to control operation of the lift system **70** by the controller **96**. In a step **206**, the controller **96** directs the lift system **70** to adjust the height of the patient support deck **38** relative to the base **34**. In a step **208**, the controller **96** directs the lift system **70** to adjust a tilt of the patient support deck **38** relative to the base **34**. In a step **210**, the controller **96** directs the lift system **70** to control articulation of the patient support surface **38**. In a step **212**, the controller **96** directs the drive system **90** to control movement of the patient support apparatus **30** over a floor surface, such as by controlling power to a powered wheel of the patient support apparatus **30**.

(57) In a step **214**, the second user interface **68** coupled to the support structure **32** is activated by the controller **96** to generate an input signal upon receiving input from the user to control an operation of the patient support apparatus **30** when the height sensor **92** detects the height of the patient support deck **38** to be above the predefined threshold or at the maximum height. Those skilled in the art appreciate that the method of controlling movement of the patient support deck **38** may include additional or alternative steps besides those shown in FIG. **6** in view of the spirit of the various examples described herein.

(58) In yet another embodiment, the patient support apparatus **30** comprises a cot. The patient support apparatus **30** further comprises a bar **110** (see FIG. **2**, shown in dotted lines) coupled to the base **34**. In some embodiments, the bar **110** could be replaced by a pedal, button, or similar design, that is easily activated by foot (and provides an associated input signal to the controller **96**). When the bar **110** is depressed, an input signal is received by the controller **96** to indicate that the patient support deck **38** should be raised. The controller **96** then generates a command signal to the lift system **70** so that the patient support deck **38** is lifted relative to the base **34** until it has reached a predefined initial height (e.g., 10 inches) as detected by the height sensor **92** (which is being monitored by the controller **96**). This function allows the caregivers to raise the patient support apparatus **30** a small amount using their feet to avoid the strain from bending or kneeling to a very low level, as cots sometimes must be very low to the ground (e.g., when patients are being loaded/unloaded).

(59) When the height sensor **92** coupled to the controller **96** detects that the patient support deck **38** is at or above the predefined initial height, the bar **110** may be deactivated (and thus rendered inoperable for any further raising of the patient support deck **38**). Thereafter, raising of the patient support deck **38** is accomplished using the user interface **66**, which may be located on the support

frame 36, patient support deck 38, or elsewhere on the patient support apparatus 30. In other words, while the user steps on the bar 110, the controller 96 continuously monitors the current height via the height sensor 92 and, as long the current height is below the predefined initial height, the controller 96 continues raising the patient support deck 38 via the lift system 70, until a comparison of the current height and the predefined initial height by the controller 96 shows that the current height meets or exceeds the predefined initial height. Once the patient support deck 38 is at or above the predefined initial height, the user interface 66 may be used, as described above, to operate the lift system 70.

(60) It will be further appreciated that the terms “include,” “includes,” and “including” have the same meaning as the terms “comprise,” “comprises,” and “comprising.” Moreover, it will be appreciated that terms such as “first,” “second,” “third,” and the like are used herein to differentiate certain structural features and components for the non-limiting, illustrative purposes of clarity and consistency.

(61) Several configurations have been discussed in the foregoing description. However, the configurations discussed herein are not intended to be exhaustive or limit the invention to any particular form. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the invention may be practiced otherwise than as specifically described.

(62) The invention is intended to be defined in the independent claims, with specific features laid out in the dependent claims, wherein the subject-matter of a claim dependent from one independent claim can also be implemented in connection with another independent claim.

Claims

1. A patient support apparatus comprising: a support structure comprising: a base, and a patient support deck coupled to the base, the patient support deck comprising a patient support surface capable of articulating to adjust positioning of a patient supported thereon; a lift system configured to adjust a height of the patient support deck relative to the base; and a user interface system coupled to the support structure and comprising an elongated member a user interface movably coupled to the elongated member and configured to be one or more of translated along the elongated member and rotated about the elongated member, the user interface configured to receive inputs from a user and comprising a gesture sensing device for monitoring motion of the user, and including a controller to control a function of the patient support apparatus based on user engagement with the user interface system: wherein the user interface system comprises a second user interface coupled to the support structure and configured to receive inputs from a user to control the function of the patient support apparatus; and wherein the user interface coupled to the elongated member is configured to be active to generate an input signal to control operation of the lift system upon receiving input from the user when a detected height of the patient support deck reaches a first threshold and to be inactive when the detected height of the patient support deck reaches a second threshold, wherein the second user interface is configured to be active to generate an input signal to control operation of the lift system upon receiving input from the user when the detected height of the patient support deck reaches the second threshold.

2. The patient support apparatus of claim 1, wherein the elongated member comprises an IV pole.

3. The patient support apparatus of claim 1, wherein the user interface is configured to receive inputs to control operation of the lift system to adjust one or more of: height of the patient support deck relative to the base; and tilt of the patient support deck relative to the base.

4. The patient support apparatus of claim 1, wherein the user interface is configured to receive inputs to control one or more of: articulation of the patient support surface relative to the base; movement of the patient support apparatus over a surface; and activation of an alarm.

5. The patient support apparatus of claim 1, wherein the user interface comprises a capacitive

sensor.

6. The patient support apparatus of claim 1, wherein the controller is configured to determine a control attribute associated with the user interface, the control attribute comprising one or more of force applied to the user interface, displacement of the user interface, velocity of the user interface, and acceleration of the user interface.
7. The patient support apparatus of claim 6, wherein the controller is configured to, based on the control attribute, control one or more of displacement of the patient support deck relative to the base, velocity in moving the patient support deck, and acceleration in moving the patient support deck.
8. The patient support apparatus of claim 1, wherein the user interface further comprises an indicator for indicating that the user interface is active to perform the function of the patient support apparatus, wherein the indicator comprises one or more of a visual indicator, a tactile indicator, and an audible indicator.
9. The patient support apparatus of claim 1, wherein the gesture sensing device comprises a camera for monitoring motion of the user relative to the user interface system.
10. A patient support apparatus comprising: a support structure comprising: a base, and a patient support deck coupled to the base, the patient support deck comprising a patient support surface capable of articulating to adjust positioning of a patient supported thereon; a lift system configured to adjust a height of the patient support deck relative to the base; and a user interface system comprising an elongated member coupled to the support structure and a first user interface movably coupled to the elongated member and configured to be one or more of translated along the elongated member and rotated about the elongated member, the first user interface configured to receive inputs from a user and comprising a gesture sensing device for monitoring motion of the user, and including a controller to control operation of the lift system when a detected height of the patient support deck reaches a first threshold and to be inactive when the detected height of the patient support deck reaches a second threshold, and wherein the user interface system comprises a second user interface coupled to the support structure and configured to be active to generate an input signal upon receiving input to control operation of the lift system when the detected height of the patient support deck reaches the second threshold.
11. The patient support apparatus of claim 10, wherein the first user interface is disposed at a position that is higher relative to a position of the second user interface.
12. The patient support apparatus of claim 10, wherein the controller is configured to determine a control attribute associated with the first user interface, the control attribute comprising one or more of force applied to the first user interface, displacement of the first user interface, velocity of the first user interface, and acceleration of the first user interface.
13. The patient support apparatus of claim 12, wherein the controller is configured to, based on the control attribute, control one or more of displacement of the patient support deck relative to the base, velocity in moving the patient support deck, and acceleration in moving the patient support deck.
14. The patient support apparatus of claim 10, wherein the first user interface is configured to receive inputs to control operation of the lift system to adjust one or more of: height of the patient support deck relative to the base; and tilt of the patient support deck relative to the base.
15. The patient support apparatus of claim 10, wherein the first user interface is configured to receive inputs to control one or more of: articulation of the patient support surface relative to the base; movement of the patient support apparatus over a surface; and activation of an alarm.
16. The patient support apparatus of claim 10, wherein the first user interface comprises a capacitive sensor interface.
17. The patient support apparatus of claim 10, wherein the user interface system further comprises an indicator for indicating that the first user interface is active to perform a function of the patient support apparatus, wherein the indicator comprises one or more of a visual indicator, a tactile

indicator, and an audible indicator.

18. The patient support apparatus of claim 10, wherein the gesture sensing device comprises a camera for monitoring motion of the user relative to the user interface system.
